

YASH VERMA

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Education

University of Michigan, Ann Arbor

Aug 2026 – May 2028

Master of Science in Computer Science and Engineering

University of Texas at San Antonio

Graduated May 2026

Bachelor of Science (Triple Major) in Computer Science, Mathematics, and Software Engineering

Overall GPA: 3.71

Technical Skills

Languages: Python, Java, C, Shell, SQL, TypeScript, JavaScript, CSS, HTML, Swift, Kotlin, R, XML, LaTeX

Developer Tools: AWS, GCP, Azure, Linux, HashiCorp, Kubernetes, Docker, Kafka, Spring Boot, Atlassian, Git, VS Code

Databases: MySQL, AWS (DynamoDB, RDS, Aurora, Neptune), SQLite, Redis, PostgreSQL, MongoDB

Experience

University of Michigan, OrderLab

Aug 2026 – Present

Graduate Research Assistant - AI Systems / AI Infrastructure

Ann Arbor, MI

- Researching and collaborating on reliable ML infrastructure for frontier LLM training on 1000+ GPU clusters, where silent distributed failures can waste millions of GPU hours per run on models scaling to 671B parameters.
- Implementing **OpGuard** (OSDI '26) bitwise-alignment debugging for production LLM training, localizing silent multi-node GPU faults within 1 iteration using bit-exact tensor checks.

University of Texas at San Antonio, CloudSys Lab

Jul 2025 – Present

Undergraduate Research Assistant

San Antonio, TX

- Advancing **KubeLLM** (IEEE Cloud Summit 2025 Best Paper) with agentic AI and a Verification Agent validating Kubernetes repairs via kubectli; achieved 100% ground-truth agreement across 40 runs on 13 benchmarks through systematic problem solving.
- Optimizing RAG pipelines (PgVector knowledge base) and multi-agent LLM orchestration for automated diagnosis and repair, reaching 95% verified task success with 10-15 second validation per task vs several minutes manually; communicated results to research teammates.

Udemy Inc.

Jun 2025 – Aug 2025

Software Engineering Intern — Design Systems Engineering Team

Austin, TX

- Deployed Django services on AWS EKS with Kubernetes and Docker for a platform serving 50M+ learners; practiced teamwork in Agile sprints via Jira, Confluence, and GitHub.
- Integrated the Figma MCP Server for agentic AI workflows, reducing setup time by 85% and improving Cursor developer outputs; communicated results and presented findings to cross-functional stakeholders.

Projects

Aegon — Agentic Control Plane for Intelligent Infrastructure | *Python, AWS, LangChain*

- Designing a 4-agent control plane for CPU/GPU/serverless, 27% lower cost and 21% higher throughput vs baseline.
- Building LLM verification with LangChain and Grafana, reducing failure recovery to under 20 seconds.

CUDA-Accelerated Transformer Inference Engine | *C++, CUDA, PyTorch*

- Building CUDA kernels for transformer attention and FFN with KV-cache reuse and fused ops for low-latency LLM inference.
- Optimizing single-GPU inference to deliver 3x higher tokens/sec than PyTorch eager on 7B-parameter models.

Publications & Leadership

Word2Vec4Kids: Interactive Challenges to Introduce Middle School Students to Word Embeddings

AAAI/EAAI 2025 | DOI: 10.1609/aaai.v39i28.35197

Association for Computing Machinery (ACM), Rowdy Creators

Aug 2023 – May 2026

Director (formerly Chief Technical Officer)

The University of Texas at San Antonio

- Demonstrated leadership through a full organizational rebrand; mentored 7 officers, hosted weekly Fortune 500 sessions that increased member participation by 40%, grew activity across 8 teams, and raised \$2,000+ in funding.